

ML Problem Solving

Linear Models, Least Squares & Classification Boundaries

Dataset Description

Consider the following 2-dimensional dataset $\mathcal{D}$ consisting of $N = 6$ training samples, where each row represents $[x_1, x_2, y]$, with inputs $x_1, x_2 \in \mathbb{R}$ and target binary label $y \in \{0, 1\}$:

$$\mathcal{D} = \begin{pmatrix} 6 & 5 & 1 \\ 5 & 6 & 0 \\ 4 & 3 & 1 \\ 3 & 4 & 0 \\ 2 & 3 & 0 \\ 3 & 2 & 1 \end{pmatrix}$$

We denote the design matrix with an explicit intercept term (column of 1s) as:

$$X = \begin{pmatrix} 1 & 6 & 5 \\ 1 & 5 & 6 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \in \mathbb{R}^{6 \times 3}, \quad y = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^6, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \in \mathbb{R}^3$$

The model prediction for an input $\mathbf{x} = [1, x_1, x_2]^T$ is given by:

$$\hat{y} = f(\mathbf{x}) = \mathbf{x}^T \beta = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

When used as a binary classifier, the predicted class label $\hat{C} \in \{0, 1\}$ is determined by thresholding at $\tau = 0.5$:

$$\hat{C}(\mathbf{x}) = \begin{cases} 1 & \text{if } f(\mathbf{x}) \geq 0.5 \\ 0 & \text{if } f(\mathbf{x}) < 0.5 \end{cases}$$

Part 1: Geometric Intuition & Data Representation

Question 1.1: Feature Space Plotting

Plot the 6 data points in the (x_1, x_2) coordinate plane: * Use distinct visual markers: filled circles $\bullet$ for class $y = 1$ and open circles/crosses $\times$ for class $y = 0$. * Label coordinates next to each point.

Question 1.2: Separability & Qualitative Inspection

1. Is this training dataset linearly separable in $\mathbb{R}^2$? If yes, sketch an intuitive linear decision boundary separating the two classes.
2. Notice the relationship between the features: along which direction do the classes appear to separate? (Look at the difference $x_1 - x_2$).

Part 2: Manual Parameter Search & Residual Sum of Squares (RSS)

The Residual Sum of Squares (RSS) is defined as:

$$\text{RSS}(\beta) = \sum_{i=1}^N (y_i - f(\mathbf{x}_i))^2 = (y - X\beta)^T (y - X\beta)$$

Question 2.1: Hypothesis Testing

As a team, select **three distinct candidate weight vectors** $\beta_a, \beta_b, \beta_c \in \mathbb{R}^3$: * **Candidate A (Baseline / Naive guess)**: E.g., setting all weights to zero or trivial values. * **Candidate B (Geometric intuition)**: A line that roughly passes through the middle between classes (hint: inspect the line $x_1 - x_2 = 0$). * **Candidate C (An aggressive or alternative slope)**: E.g., heavily weighting only one feature.

Record your chosen vectors: * $\beta_a = [\beta_{0,a}, \beta_{1,a}, \beta_{2,a}]^T = [_, _, _]^T$ * $\beta_b = [\beta_{0,b}, \beta_{1,b}, \beta_{2,b}]^T = [_, _, _]^T$ * $\beta_c = [\beta_{0,c}, \beta_{1,c}, \beta_{2,c}]^T = [_, _, _]^T$

Question 2.2: RSS Computation Table

Complete the following calculation table for each candidate β . Compute predictions $\hat{y}^{(i)} = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2}$, the residual $e_i = y_i - \hat{y}^{(i)}$, and squared residual e_i^2 :

Sample i	$(x_{i,1}, x_{i,2})$	True y_i	$\hat{y}_a$	$(y_i - \hat{y}_a)^2$	$\hat{y}_b$	$(y_i - \hat{y}_b)^2$	$\hat{y}_c$	$(y_i - \hat{y}_c)^2$
1	(6, 5)	1						
2	(5, 6)	0						
3	(4, 3)	1						
4	(3, 4)	0						
5	(2, 3)	0						
6	(3, 2)	1						
Total	—	—	—	RSS_a =	—	RSS_b =	—	RSS_c =
RSS								

Question 2.3: Decision Boundary Equations

For each of your 3 models, write the explicit equation of the decision boundary corresponding to the classification threshold $\hat{y} = 0.5$:

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0.5 \implies x_2 = m \cdot x_1 + c$$

1. Solve for slope m and intercept c for models a, b , and c . 2. Overlay these three decision lines on your plot from Question 1.1. 3. Count the classification training errors (0-1 loss) for each candidate. Does the vector with the lowest RSS necessarily yield the lowest classification error? Explain why or why not.

Part 3: Step-by-Step Exact Numerical Solution

Now, your team will compute the exact OLS solution $\hat{\beta} = (X^T X)^{-1} X^T y$ manually or with basic arithmetic verification.

Question 3.1: Compute the Gram Matrix $X^T X$

Given:

$$X = \begin{pmatrix} 1 & 6 & 5 \\ 1 & 5 & 6 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

1. Compute each element of $G = X^T X \in \mathbb{R}^{3 \times 3}$
2. Compute the vector $X^T y \in \mathbb{R}^3$

Question 3.2: Invert $X^T X$ and Solve for $\hat{\beta}$

1. Calculate the determinant $\det(X^T X)$.
2. Calculate the inverse matrix $(X^T X)^{-1}$.
3. Multiply $(X^T X)^{-1}(X^T y)$ to obtain the optimal weight vector $\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2]^T$.
4. What is the optimal decision boundary line $x_2 = m^* x_1 + c^*$?
5. Compute the optimal RSS($\hat{\beta}$). Verify that $\text{RSS}(\hat{\beta}) \leq \min(\text{RSS}_a, \text{RSS}_b, \text{RSS}_c)$ from Part 2.